Literature Review: Voice Spoofing Detection

Advanced Topics in Machine Learning • Stage 1

1. Abstract

The evolution of modern Text-to-Speech (TTS) and Voice Conversion (VC) systems has led to an unprecedented improvement in synthetic speech quality. This trend is driven primarily by the shift from traditional vocoders to neural network-based codecs (Neural Codecs) such as EnCodec, DAC, and Speech Tokenizer, which generate audio signals nearly indistinguishable from human speech. This review analyzes the current challenges in Automatic Speaker Verification (ASV) and Anti-Spoofing systems, emphasizing the Generalization Gap against unknown (Zero-day) attacks. The document examines nine key papers covering novel datasets, feature extraction methods, complex classification architectures, and Anomaly Detection paradigms, aiming to establish a theoretical and empirical foundation for developing a robust classifier.

2. Introduction: The Generalization Gap and Empirical Risk

The core challenge in Audio Anti-Spoofing research is no longer building a model that successfully identifies known attacks, but rather constructing a system capable of generalizing to Out-of-Distribution (OOD) attacks. Statistical learning theory can mathematically formalize this challenge.

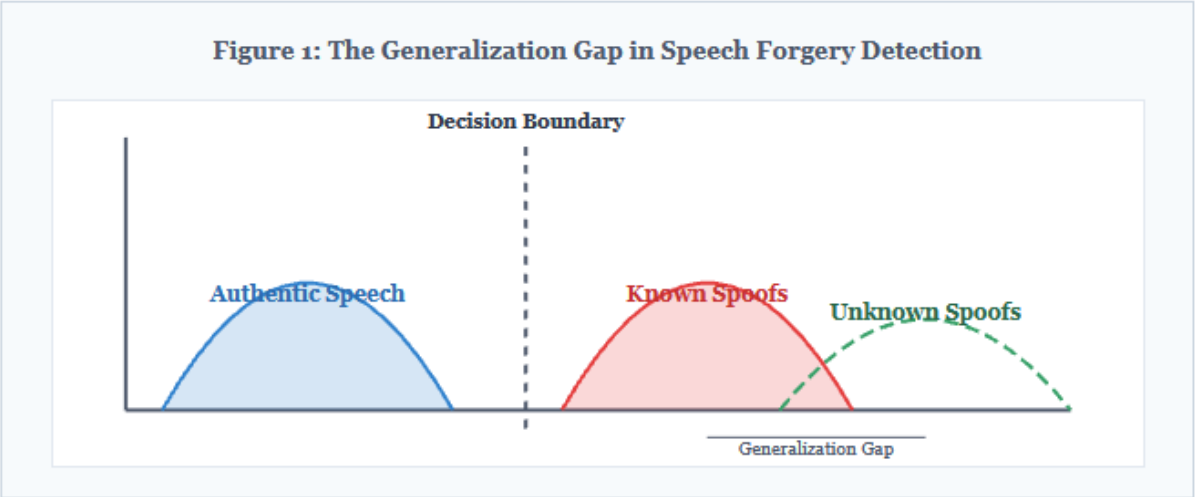
In standard training, a machine learning model aims to minimize the **Empirical Risk** over a training set comprising authentic samples and known synthetic anomalies. The empirical risk is defined as:

$$R_{emp}(h) = (1/n) \sum_{i=1}^{n} L(h(x_i), y_i)$$

However, because the landscape of potential attacks is constantly expanding (e.g., with the advent of models like VALL-E), the training distribution never truly represents the **True Risk**, defined as the expected error over the true data distribution:

$$R(h) = E_{(x,y)\sim P} [L(h(x), y)]$$

The difference between true and empirical risk is the "Generalization Gap." Models achieving near-zero error on legacy spoofing datasets often fail catastrophically when exposed to the subtle artifacts of Neural Codecs because their decision boundaries overly conform (overfit) to the digital signatures of older methods.



Visual representation of how minimizing empirical risk on known spoofing distributions (red) fails to capture novel, unseen attack distributions (green), highlighting the generalization gap.

3. Datasets and Evaluation Frameworks

To examine and mitigate the generalization gap, the research community has developed datasets that more accurately represent real-world threats.

3.1 ASVspoof 5 and Evaluation Metrics

The fifth iteration of the **ASVspoof 5** challenge [1] represents a significant leap by incorporating crowdsourced data collected under highly variable acoustic conditions. Crucially, the dataset introduces adversarial attacks (such as Malafide and Malacopula filters) specifically designed to fool detection and automatic speaker verification (ASV) systems. Furthermore, the challenge emphasizes transitioning from the standard Equal Error Rate (EER) to a tailored metric called **a-DCF (architecture-agnostic Detection Cost Function).** This metric weighs error costs according to different use cases, heavily penalizing models that produce high-confidence errors on zero-day deepfakes. The severity of this generalization gap is evident in the baseline results: state-of-the-art models like AASIST experienced severe performance degradation on ASVspoof 5, yielding an EER of 29.12%, whereas the leading open systems adapted to these conditions achieved an EER of 2.59% [1].

3.2 CodecFake: Countering Neural Codecs

While ASVspoof provides a broad benchmark, the **CodecFake** dataset [2] was developed as a direct response to the vulnerability of verification systems against neural codec-based speech synthesis. Traditional models fail to detect deepfakes generated by VALL-E and similar architectures because Vector Quantization artifacts differ fundamentally from those of classic vocoders. The dataset utilizes audio reconstruction via codecs to generate samples

that preserve the exact signatures of these coding methods. Training on CodecFake has empirically demonstrated the ability to drop the EER to as low as 0.4% when defending against state-of-the-art VALL-E attacks.

3.3 Natural Baseline (M-AILABS)

To implement anomaly detection paradigms, a precise characterization of the "normal speech" distribution is required. Using studio-quality recording datasets like M-AILABS allows models to learn natural acoustic and phonetic variance without the influence of channel effects that could create spurious overlaps with synthesis artifacts.

4. Feature Representation and Processing

Classification capabilities depend heavily on the quality of representations extracted from the audio signal. The literature highlights two main pathways: advanced spectral features and deep self-supervised representation learning.

4.1 Spectral Features: Constant Q Transform (CQCC)

The traditional Short-Time Fourier Transform (STFT) utilizes a fixed time-frequency resolution, which often masks high-frequency artifacts caused by synthesis model flaws (e.g., spectral leakage). To resolve this, **CQCC (Constant Q Cepstral Coefficients)** [3] was introduced, based on the Constant Q Transform (CQT). In CQT, the frequency-to-bandwidth ratio remains constant: $Q = f_k / \Delta f_k$. This method provides high frequency resolution at lower frequencies (vital for human speech) and high temporal resolution at high frequencies (vital for digital artifacts). Applying Cepstral Analysis to the CQT spectrum captures non-linear distortions that are difficult to detect with standard MFCCs.

4.2 Self-Supervised Learning (SSL)

Modern approaches utilize foundation models pre-trained on massive amounts of unlabeled speech directly from the raw waveform.

wav2vec 2.0: A model [4] that uses a convolutional network to represent the raw waveform as a latent space sequence. Portions of the representations are masked, and a Transformer model must reconstruct them by solving a contrastive task against quantized representations. The contrastive loss function is defined as:

$$L_m = -\log\left(\exp(\mathrm{sim}(c_t, q_t)/\kappa) \,/\, \sum_{q \in Q} \exp(\mathrm{sim}(c_t, q)/\kappa) \right)$$

where c_t is the context vector of the masked layer, q_t is the true target representation, sim is a cosine similarity function, and κ is the temperature parameter. This process compels the model to learn deep acoustic structures distinguishing natural speech from artificial signals.

WavLM: While powerful, wav2vec 2.0 focuses primarily on speech content (ASR). **WavLM** [5] extends this paradigm by simulating noisy environments and creating overlapped speech

during pre-training on 94,000 hours of audio. Alongside predicting masked portions, the model must perform denoising, forcing it to learn rich representations of **speaker identity** and paralinguistics. Consequently, WavLM features have proven in the literature to be exceptionally robust representations for anomaly detection and identifying voice signatures generated by synthetic models.

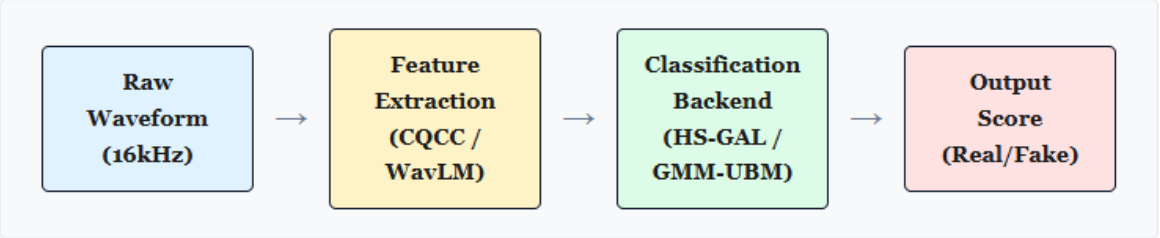
5. Modern Classification Architectures

Above the feature representation layer, specialized classification architectures are deployed to detect malicious patterns.

5.1 AASIST (Integrated Spectro-Temporal Graph Attention Networks)

Standard CNNs applied to spectrograms face limitations in capturing long-range dependencies between the time and frequency axes. The **AASIST** framework [6] resolves this by modeling audio as a graph. The architecture employs **Heterogeneous Stacking Graph Attention Layers (HS-GAL),** which separately and subsequently jointly model artifacts in the spectral and temporal domains. Utilizing a competitive Max Graph Operation (MGO), the network isolates specific regions in the signal containing forensic cues. Empirically, this approach improves upon state-of-the-art baselines by over 20% relative in terms of pooled min t-DCF, achieving an EER of 0.83% in optimal configurations.

Figure 2: Advanced Audio Anti-Spoofing Architecture



Typical pipeline: Raw audio is processed via handcrafted features (CQCC) or SSL models (WavLM) and fed into specialized backends.

5.2 Hybrid Classification (CNN + GMM-UBM)

In an alternative, lightweight approach, classical statistical models can be integrated with deep networks [7]. Izrailov et al. (2026) demonstrate a hybrid architecture where acoustic features, specifically Mel-Frequency Cepstral Coefficients (MFCC) and Perceptual Linear Prediction (PLP)- are evaluated against a Universal Background Model (UBM). The adapted Gaussian Mixture Model (GMM) parameters serve as a sparse statistical representation of the voice, which is then fed into a compact Convolutional Neural Network (CNN). This hybrid pipeline reduces the EER to 4.79% on standard logical access datasets, maintaining

high robustness while significantly reducing the computational overhead compared to heavy Transformer models.

6. Generalization Paradigms and Anomaly Detection

To overcome the true risk issue described in the introduction, researchers have developed methods that decouple the model from relying on pre-labeled spoofing data.

6.1 One-Class Learning (Deep SVDD)

Traditional binary classification networks attempt to draw a boundary between two distributions. The One-Class approach is based on the premise that authentic speech is stable and bounded, whereas spoofing attempts are unpredictable deviations. Yin & Lei (2026) integrate deep wav2vec 2.0 representations fused with classical LFCC (Linear Frequency Cepstral Coefficients) features into a **Deep SVDD (Support Vector Data Description)** architecture [8], training the system to generate the tightest possible hypersphere strictly around authentic training data.

The objective function minimizes the average distance of real speech samples from the center c in the latent space:

$$\min_W (1/n) \sum_{i=1}^{n} \| f(x_i; W) - c \|^2 + (\lambda/2) \| W \|^2$$

Consequently, any recording (even from a novel Codec attack) deviating from the human hypersphere by a distance greater than the defined radius will be classified as a spoof, regardless of how it was generated. This purely anomaly-based modeling achieved a highly competitive 6.89% EER against entirely unknown attacks on the ASVspoof 2021 DF dataset.

6.2 Enhanced Forensic Trace Generation

An active method for improving generalization involves using audio generators to amplify spoofing signatures [9]. Wu et al. (2026) utilize a VQ-VAE model as a generator to reconstruct speech spectrograms, which structurally amplifies vocoder distortions and frequency inconsistencies. This generation is supervised by a Knowledge-Driven Generation Optimization strategy guided by three components: True Speech Consistency Loss, Triplet Loss, and a Positive Sample Constraint [9]. This forces the generator to maximize the spatial separation between authentic speech and reconstructed deepfakes, empirically reducing the EER of standard spectral features (like LPS) from 22.64% down to 14.04% [9]. Calculating the reconstruction difference allows the classifier to ignore environmental variables and focus solely on the amplified forensic traces of digital forgery.

7. Conclusion and Baseline Transition

This review elucidates that the complexity of voice spoofing detection stems from critical generalization gaps, necessitating a combination of rigorous evaluation frameworks (such as CodecFake and ASVspoof 5), advanced representation learning (WavLM/wav2vec), and anomaly modeling (One-Class).

To quantify and empirically validate these theories within this project, Stage 2 will commence by establishing a simple Baseline model. This model will rely on less complex feature extraction- **Mel-frequency cepstral coefficients (MFCC)** combined with a basic statistical classifier (Logistic Regression). This traditional approach is not expected to solve the Zero-day problem of Neural Codecs; rather, its purpose is to demonstrate and prove its failure on novel datasets. Establishing this baseline will serve as an essential quantitative reference point to justify the future integration of WavLM representations and One-Class classification architectures on the path to fully addressing the true risk.

References

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